

# Where a quantum or generative advantage can, and cannot, live

The structure that makes SQD classically *verifiable* is the structure that makes it classically *constructible*.  
Advantage survives only where that equivalence breaks.

## Classically verifiable & simulable

### CLASSICAL WINS

**Task:** molecular ground-state SQD

- SHCI / CIPSI / DMRG / AFQMC reach the ~1.3 mHa consensus at lower cost
- LUCJ circuits dequantized on a laptop [16]; random+recovery matches
- exponential even for a perfect sampler in delocalized states [17]

*The generative advantage is dequantized along with the sampler.*

## Classical signal degrades

### OPEN / CONDITIONAL

**Where the cheap prior fails:**

- high noise → starved valid-shot budget; symmetry-valid generators waste nothing
- strong multireference → HF-based perturbative ranking misfires
- amortization across a PES / chemical series (train once)

*The empirical target of \$6–\$7 (GFlowNet).*

## Provably hard classically

### ADVANTAGE = THEOREM

**Task:** learning from experiments

- two-copy / quantum-memory learning: exponential separation, **unconditional** (a theorem, not a conjecture) [11,12]
- cannot be dequantized

**Caveats:** tasks are contrived; fragile on NISQ hardware [53]; useful reach = device characterization

*The highest-value open bridge to chemistry.*

cheap classical construction & verification

classical construction provably expensive

→ aim generative & quantum methods where classical proof is NOT cheap →